Alchemy of Things

A framework for recombination under constraint

First edition · 2026 · Takura Changwa

A note on rigour. This is a first edition. It is unpublished and unreviewed.

Maybe it can be the first of many subjects to be tested by its own methods.

Several of the concepts I borrow here come from fields I do not practise, and no part of this document has been read by a specialist in any of them. Where it reaches into physics, chemistry, engineering or philosophy, I am reading outside my training. I offer those parallels as resonances, not as proofs, and maybe they are worth holding loosely.

This is a proposition to be tested, not a finding. I am not simply open to correction. I am asking for it.

Seeing Connections

I am an architect. I have been practising for about twenty years.

Architects are trained to see buildings. I have realised that I tend to see something else. I see potential connections.

I don't entirely accept definitions. The moment we define a thing, we tend to give it a use. A floor tile becomes something that belongs on a floor. A tyre becomes something that belongs on a car. A building becomes a building, rather than a collection of materials, spaces, histories and possibilities.

Definitions are useful, but they can be plagued by set preconceptions. They tell us what a thing is, and quietly begin to tell us what it can be. When we loosen those definitions, the thing enters a kind of state of potential. It becomes an active ingredient, available to new connections, new relationships and new propositions.

I have used a floor tile as a kitchen countertop, not as an aesthetic experiment but as a response to a budgetary constraint: it was significantly cheaper than conventional stone countertop surfaces and therefore became a viable solution within the limits of what I could afford at the time. Before that, I watched my mother cut a strip from a used tyre and make a hinge of it. The material did not change. Its relationship to the things around it changed.

I did not learn this at school. I learned it watching someone solve a problem with what was already there.

A crescent moon can remain a crescent moon, but it can also become the starting point for a painting about grief, longing or joy. A piece of discarded timber can become a bench. A vacant building can become a community resource. A disagreement can become information about how a relationship needs to change.

I see things less as objects with prescribed uses and more as ingredients with potential relationships. The interesting question is not only: what is this? It is also:

What else could it be?

What could it connect to?

What could happen if I put it somewhere it does not normally belong?

This is what I have come to call alchemy. Not because anything supernatural happens, but because the transformation often begins with a change in relationship rather than a change in material. The thing remains what it was. Our understanding of what it can do changes.

What follows is an attempt to describe this way of seeing without making it dependent on its author. If the three laws hold, they should hold for a farmer and a treasurer as readily as for an architect. If they hold only for the person who wrote them, they are not laws. They are a portrait.

I The Thesis

What the framework claims, who it is for, and where the three laws come from.

This framework begins with a simple observation: we tend to define things by what we believe they are, and in doing so we often define what they are for.

A floor tile becomes something for a floor. A tyre belongs to a car. A building becomes a building. Definitions are useful. They allow us to communicate, classify and organise the world. But they can also become boundaries. Once a thing has been given a category, its other possibilities become harder to see.

I am interested in what happens when those boundaries are loosened.

Any domain can be read as a collection of ingredients and relationships rather than a collection of fixed categories. Once read this way, things can be recombined across boundaries that specialists, institutions and habits often treat as walls.

This is not a method belonging to designers. It is what anyone does who makes, maintains or runs anything.

A farmer deciding what to plant beside what. A nurse reorganising a ward round. A teacher reordering a syllabus. A shopkeeper arranging stock. A treasurer arranging money. A parent running a household on a fixed income.

None of them create the world they are working with from nothing. They work with what is already there.

They notice ingredients. They establish relationships between them. They remove some things, introduce others, change the sequence, alter the conditions, and see what emerges.

The quality of an arrangement depends not only on the ingredients, but on how they relate.

This has an interesting parallel with Carlo Rovelli's relational view of reality. In his interpretation of quantum mechanics, physical properties are not understood as existing entirely in isolation, but in relation to something else. More broadly, Rovelli describes reality not as a collection of independently existing things, but as a network of relations.

I am not proposing that the Alchemy of Things is a theory of physics. The connection is more modest, and perhaps more useful: it offers a way of thinking about things in which relationships are not secondary to the things themselves.

Perhaps things are less fixed than the categories we give them. Perhaps what something can become is partly determined by what it is connected to.

Where the laws come from

Everything above is observation. Three things in it keep recurring, and I want to state them as laws because I think they hold beyond the examples I have used.

The first comes from the fact that none of the people described above are making anything out of nothing. They are arranging what is already in front of them. If that is true generally, then everything made is assembled from things that already existed.

The second comes from the accumulation. Arrangements do not hold themselves. They have to be maintained, and when the maintaining stops, they come apart. If that is true generally, then everything organised requires flows through it to remain organised.

The third comes from the relationships. If what a thing can become depends on what it is connected to, then how well it works is not a property it holds by itself. It is a question of how it sits with everything around it.

First: everything made is assembled from things that already existed.

Second: everything organised requires flows through it to remain organised.

Third: something works to the degree that it fits the flows, conditions and relationships around it.

The rest of this framework is an attempt to understand what they mean, how they can be used, and where they might fail.

II Recombination

The first law. Nothing is made from nothing.

Nothing is made from nothing. All novelty is the rearrangement of existing components across contexts.

Every made thing is assembled from things that already existed, recursively, all the way down. A tool from earlier tools, a recipe from earlier recipes, a law from earlier laws. And because the number of possible combinations grows faster than the number of parts, whoever works across several fields has more available moves than whoever works inside one.

The work is not inventing ingredients. It is holding an unusually large and deeply understood pantry, and cooking across cuisines.

III Classification

What the first law requires before anything can be combined: knowing what you actually have.

If everything made is assembled from what already exists, then the first practical question is what the parts are, and which of them can be changed. That is what this classification is for. It sorts whatever is in front of you into four kinds.

Core elements

Irreducible inputs with fixed properties. Cannot be argued with, only worked with or against.

Diagnostic — What is true here whether or not anyone likes it?

Catalysts

Not consumed in the reaction, but they change what reactions are possible or how fast they occur.

Diagnostic — What changes what is possible without being used up?

Processes

Transformations over time. Ordered, with conditions of failure.

Diagnostic — What is the sequence, and where does it break?

Outputs

What exists at completion. Outputs re-enter the system as core elements for later work.

Diagnostic — What is now true in the world that was not before?

Core elements: the climate of a place, the strength of a material, the money actually in the account, the law as written, the hours in a week, the soil, the diagnosis, what a colleague can genuinely do. Catalysts: a tool, a relationship, an introduction, a reframing of the question, a precedent borrowed from a field with no connection to yours. Processes: a design stage, a harvest, a ward round, a rehearsal, a negotiation, a hiring round. Outputs: a finished building, a trained team, a crop sold, a passed policy, a working piece of software, a garden that feeds people.

The fourth class is where the first law becomes visible over time. An output does not sit outside the system once it is finished. It goes back in as an ingredient, which is why a completed building, a trained team or a working piece of software becomes a core element for whatever comes next.

IV Reaction Types

What actually happens when two elements meet, since combination is not one thing.

The first law says that things get made by combining what exists. It does not say that every combination produces something. Several different things can happen when two elements meet, and it matters which, because otherwise every combination gets described as a success and the most common outcome goes unnamed.

Synthesis

Two elements form a third with properties neither had.

Signature — The output cannot be decomposed back into its inputs.

Catalysis

One element accelerates a reaction without entering it.

Signature — Remove it and the reaction still happens, far slower or not at all.

Inhibition

One element blocks a reaction that would otherwise occur.

Signature — Progress stops and nobody can name the blocker. Usually a person or an unexamined assumption.

Decomposition

A complex output breaks into components usable elsewhere.

Signature — The project fails but the parts survive.

Inert pairing

Two elements coexist without reacting.

Signature — Meetings happen. Nothing changes.

Inert pairing is the most common outcome in practice and the one people are worst at admitting to. Assume it is the default until evidence says otherwise.

Decomposition is worth holding on to, because the first law makes it a result rather than a loss. When a project fails, the parts do not disappear. They return to the pantry, and they are usually better understood than they were before.

V The Precondition

The limit of the first law, and of whoever uses it.

An alchemist can only work with what they fully understand.

Recombination across domains is powerful only where the understanding of each ingredient is deep. Shallow recombination produces analogies that sound profound and collapse on contact with a specialist. Depth in the source material is the only difference between insight and pastiche.

This imposes a hard ceiling on how many domains can be held at working depth at once. Fewer ingredients understood deeply will always outperform more ingredients understood partially.

The framework does not license breadth. It constrains it.

VI Metabolism

The second law. Nothing stays in order for free.

Nothing stays in order for free. Everything that persists requires something to keep passing through it.

A living body requires food, oxygen and water. A tree requires sunlight, water and nutrients. A garden requires water, nutrients, labour and time. A city requires energy, food, money, maintenance, people and information.

The same is true of things we do not normally think of as living systems. A building requires maintenance. A business requires revenue and labour. A community requires participation. A project requires attention.

These flows are what keep a system in the state we recognise as functioning. Stop the flows and the system comes apart.

Every solution therefore has a metabolism: an ongoing set of inputs, exchanges and maintenance required to keep it working.

The operating questions are:

What has to keep moving for this to continue working?

Who provides it?

What does it cost?

What happens when that flow stops?

A solution that depends indefinitely on an invisible person, an unaffordable resource or an unsustainable amount of maintenance may have solved the immediate problem while creating another one.

Name the running cost and name who bears it in year five. This is the difference between a design and a debt.

VII Accumulation and Return

What persistence hides, and how buried ingredients are recovered.

As arrangements accumulate, however, they develop layers upon layers of complexity. Those layers can eventually obscure the base components from which they were made. We inherit systems, categories, procedures and assumptions without necessarily remembering why they were assembled that way in the first place.

Part of the process must therefore involve going back: recovering ingredients that have disappeared beneath accumulated complexity and making them available for reconsideration.

But this is not a simple cycle. We do not return to the same point. What has happened in between has changed us, the ingredients and the relationships between them.

There is a useful resonance here with adrienne maree brown's *Emergent Strategy*: change is relational, adaptive and iterative. Larger patterns emerge through interactions between smaller parts, and those emergent patterns alter the conditions for what happens next.

There is also a useful resonance with the Akan concept of Sankofa: returning to retrieve from the past what remains valuable in order to move forward. The return is not an attempt to recreate the past. It is a way of recovering what has been forgotten and allowing it to enter into new relationships.

Learning and unlearning therefore belong to the same movement. Learning builds relationships and layers of understanding. Unlearning allows us to question relationships that have become so fixed that we mistake them for the ingredients themselves.

We return, but we return changed. And what we retrieve returns to a world that has changed with us.

VIII Impedance Matching

The third law. Fit does the work, not force and not absence.

Fit does the work. Not force, and not absence.

Energy transfers usefully only where two impedances match. Impedance is the engineer's word for how much a thing resists what passes through it, and the same idea has a name in every trade: gearing, sizing, pitch, fit, tolerance. Every one of them describes the same search for the point where two things meet without either being wasted.

Offer no resistance at all and nothing is carried: the flow passes as if the thing were not there, or tears through and destroys what fed it. Offer total resistance and nothing passes either. Everything useful happens between the two, at the point of match. In electrical engineering this is the maximum power transfer theorem, and it is not a metaphor there.

A loudspeaker is not the absence of resistance to air. It is a device matched to the impedance of air, which is why it can move air at all. A tree is not a low impedance object in a landscape. It is an enormous obstruction, matched so precisely to its flows of light, water and wind that it participates in them instead of fighting them.

The goal is not the absence of the object. The goal is the object matched to its flows.

The operating question: what is flowing here, and is this thing matched to it? Not how small its footprint is. Not how rounded its edge.

IX Valence

What makes two things fit, and what makes the classification predictive.

Classification tells you what you have. It does not tell you what will combine. For that we need the third law, because two things fit or they do not, and what determines the fit is what each of them is missing.

Elements bond according to what they are missing. This, not the table itself, is where the chemistry holds.

Each element carries a valence: the specific capacity it lacks and the specific capacity it offers.

A tight budget

Offers — Forcing function, clarity, the elimination of vanity

Lacks — Range, quality of material

Bonds with — High ingenuity, low capital methods

Deep expertise in one field

Offers — Depth, judgment, credibility

Lacks — Outside reference, transferable framing

Bonds with — A collaborator from an unrelated domain

An underused space or asset

Offers — Capacity, position, a cost already paid

Lacks — Purpose, upkeep, the people to fill it

Bonds with — A group that needs room and holds none

Volunteer labour

Offers — Numbers, goodwill, low cost

Lacks — Continuity, specialised skill

Bonds with — Simple repeatable tasks, clear scheduling

A long attention span

Offers — Depth, patience, compound returns

Lacks — Speed, breadth, early feedback

Bonds with — Work whose value only appears late

Populate this to twenty rows across the real practice. Once it exists, the framework becomes predictive: scan for elements whose valences complete each other and have not yet been combined.

The periodic table is not admired for being tidy. In 1871 Mendeleev looked at the gaps in his table and predicted two elements nobody had seen, along with their properties. Gallium was found in 1875 and germanium in 1886, and both matched. A classification that only describes completed work is a filing cabinet.

A thing is also what it can enter into relation with

Use a tile as a counter, as a weight, or as one piece in a pattern, and it is the same tile each time. Nothing has happened to the material. What changes is the field of relations through which it is met. Its identity as a useful thing was never a property held in isolation.

This is where James J. Gibson's concept of affordance becomes useful. For Gibson, what an environment affords is neither simply a property of the object nor something invented entirely by the observer. It exists in relation to the capacities of the one encountering it. A surface may afford walking to one body, climbing to another, shelter to another. The physical thing remains, but different relationships make different possibilities available.

The floor tile used as a countertop is therefore not simply a case of misusing a tile. Its hardness, flatness, durability and cleanability were already present. The budgetary constraint, the requirements of the kitchen and the willingness to release the tile from the category flooring allowed another affordance to become useful.

A parallel exists in relational readings of physical systems, particularly Carlo Rovelli's Relational Quantum Mechanics, where properties attributable to a physical system are understood relative to interactions with another system rather than as an exhaustive set of properties possessed independently of all relations.

That parallel is worth holding, and worth holding loosely.

Gibson is speaking about perception and action. Rovelli is making a specific claim about quantum mechanics. Neither proves the Alchemy of Things.

What they provide are independently developed examples of a proposition important here: a thing cannot always be fully understood by examining it in isolation from what it can enter into relation with.

One caution follows from it. If identity were purely perspectival, nothing could be carried from one person to another, and the transferability test in XIV would be empty rather than demanding. Relations have to be able to hold across observers.

A tile that becomes a counter for one person only has not been recombined. It has been imagined.

X The Ladder

How far an idea travels, and why the third law decides whether it reaches the end.

An idea has to travel before it becomes anything, and the third law describes the last stage of that journey. A thing spreads on its own when it is matched well enough to the conditions around it that nobody has to keep pushing it.

Raw idea

An association between elements not conventionally combined.

Gate — Survives being stated plainly to a competent skeptic.

Where it dies — Never leaves the voice memo.

Prototype

The idea meets one real constraint: material, budget, code, or a user.

Gate — The constraint reveals what the idea is rather than killing it.

Where it dies — The constraint is avoided, so the idea stays perfect and untested.

Product

Resolved into something usable outside the conditions that made it.

Gate — Works when its author is not in the room.

Where it dies — Requires its author permanently.

Culture

Outlives its original context and changes how other things get made.

Gate — Impedance low enough to propagate without promotion.

Where it dies — Never reaches product, so the question never arises.

Most ideas die at the first gate, and they die from being protected. An idea that has never met a constraint stays perfect, and perfection is indistinguishable from untested. Any use of this framework should therefore be biased toward early contact with something real: a budget, a material, a deadline, a person who has to live with the result.

Any version that permits more time refining the framework is working against its author.

XI Evaluation

Where judgement enters, and the one criterion that comes straight out of the second law.

The three laws describe how made things behave. They do not tell anyone whether something should be made. That decision needs criteria, and these six are written so that the answer can be no. The last of them is the second law restated as a question: if order requires flows, then the running cost is part of the design and not a detail to be settled later.

Value

Name the specific person or group whose capacity increases, and by how much. Not the world.

Fails when — The value can only be described in adjectives.

Synergy

Identify the property of the output that neither input possessed.

Fails when — The output is both things next to each other.

Autonomy

Name the new dependency this creates. Every solution creates one.

Fails when — The dependency created is on one person, indefinitely.

Risk

Name the weakest assumption, what fails first when it is wrong, and how it would show early.

Fails when — The failure mode is unnamed, or the answer is that it should be fine.

Ethics

Identify who bears the cost of failure, and whether they had a say in the attempt.

Fails when — Those carrying the downside are not those making the decision.

Metabolism

State the running cost of holding this output in its ordered state, and who pays it in year five.

Fails when — It only works while one person keeps maintaining it.

Any single failure is a stop condition until it is addressed or consciously accepted in writing.

XII Operation

The procedure. Run forward on a problem, and backward on work already finished.

Everything up to here is description. This is the part that gets used. The forward move takes the laws in order: find the parts, find what they are missing, find the fit. The third step is where the accumulation and return described in VII becomes practical.

The forward move

1. State the gap. What capacity is missing. If the problem statement contains a solution, restart.
2. Inventory core elements. Be ruthless. Most things called fixed are preferences; most things called flexible are physics.
3. Look for the ingredients that have been buried. Some of what appears fixed is not a core element at all but an arrangement that hardened at some point and was never questioned. Return those to the pantry before going further.
4. Map valences. What each element offers and lacks.
5. Search for the match. Scan for elements whose valences complete each other, including from outside the domain. This is the alchemy step and the only one where the method is the differentiator.
6. Predict the reaction type. Write the prediction down before starting.
7. Run the process. Log what actually happened, not the version fit for presentation.
8. Score against all six criteria.
9. Return the output to the pantry as a core element with its valences documented.

The reverse move

Take a completed project. Decompose it into core elements, catalysts and processes. Ask which single element, removed, would have collapsed the whole thing.

That element is the method, exposed.

Run this on five completed projects. The element recurring across all five is the real framework, found rather than declared.

XIII Pillars

The limit of the whole method. These are a choice, not a consequence of the laws.

I want to be clear about something before stating these. The three laws describe how made things behave. They do not say what should be made, and no amount of working with them will produce a purpose. These three pillars are a choice I have made about what the framework is for. I am stating them as a choice rather than presenting them as something the laws require.

Reduce inefficiency

Operating question — Does this remove friction currently tolerated as normal?

Uplift society

Operating question — Does this increase what specific people can do, not what they can buy?

Restore environment

Operating question — Does this leave its site, material stream, or system in better condition than it found it?

Society and environment are not separable, so the third pillar binds as tightly as the second.

These three will collide on real projects, and they collide hardest on high value work. When they do, the ranking used to make the call gets stated in writing at the time of the decision.

An unstated ranking is how principles quietly become marketing.

XIV Falsification

How to tell whether the framework is working or only agreeing.

A framework that explains every success and no failure is not a theory. It is an identity.

Predictive

Passes if — The valence map identifies a combination not previously considered, it is pursued, and it works.

Fails if — It only explains combinations already made.

Discriminating

Passes if — It correctly identifies an exciting project as a bad combination, and the project is dropped.

Fails if — It has never said no.

Transferable

Passes if — Someone else uses it and reaches a useful conclusion without interpretation.

Fails if — It only works when its author explains it.

Post mortem

Passes if — Applied to a failure, it names the specific reason.

Fails if — It says the failure was insufficiently alchemical.

Six month review. If the framework has not once refused something wanted, it is not functioning as a framework. It is functioning as a mirror.

XV The Warning Inside the Metaphor

Why the name is the right one, and what it warns against.

Alchemy is the right name for this method. Dissolve and recombine. Its practical operations gave us metallurgy, distillation, and eventually chemistry itself.

It is also a long example of a system that was internally coherent, symbolically beautiful, explained everything in its own terms, attracted serious intellects for over a thousand years, and produced almost no reliable knowledge until it was forced to submit to external verification.

Both are true. Keep the metaphor and keep the warning bolted to it.

What turned alchemy into chemistry was not better symbolism. It was the willingness to be measured.

XVI A Request for Testing

What is asked of the reader, stated in the framework's own terms.

This framework cannot test itself.

Its propositions require application, comparison, disagreement and revision. They need to be tested by people working in different fields, under different constraints, with different kinds of knowledge. I need to know where they work, where they fail, what they reveal and what they obscure.

So I am not simply open to review. I am asking for it.

There is a term in IV for the response I am most concerned about receiving. Inert pairing: two elements sit beside each other and nothing happens. If you read this, agree with it, and put it down, that is an inert pairing, and nothing has moved in either direction. The third law explains why. Offer no resistance at all and nothing is carried. Agreement is zero resistance. It feels like something has been transmitted, and nothing has.

So what I am asking for is a match. Enough resistance to move something. Tell me where the framework does not hold, where a law does not apply in your field, where the language gets in the way of the idea, and where you have seen the opposite of what I have described.

If you use the framework, tell me what happened. Share what worked, share what failed, share the connections it helped you see and those it failed to reveal. Most importantly, share your disagreements.

A framework becomes more credible not because people agree with it, but because it can withstand people who do not.

It must also be capable of rethinking its own position. If evidence contradicts a proposition, the proposition must be reconsidered. If an assumption fails, it must be changed. If a law repeatedly fails under conditions where it should hold, the law must be rewritten or abandoned.

The ability to be wrong is not a weakness of the framework. It is a requirement. Knowing what does not work is knowledge, and through knowing what not to do we gain more evidence about what we might try.

Failure therefore becomes an ingredient.

Disagreement becomes an ingredient.

Contradiction becomes an ingredient.

The framework itself must remain available for recombination. Its authorship should therefore become increasingly collective: tested by different people, in different contexts, with results that can be compared, challenged and incorporated.

The aim is not to protect the framework. The aim is to find out whether it survives contact with the world.

XVII References

Where the borrowed concepts come from, and what has not been verified.

Sources for the concepts borrowed here. Several are drawn from fields the author does not practise, and none of this document has been read by a specialist in any of them.

Recombination

W. Brian Arthur, *The Nature of Technology: What It Is and How It Evolves*. Free Press, 2009.

Combinatorial growth

Martin L. Weitzman, *Recombinant Growth*. Quarterly Journal of Economics 113 (2), 1998, 331 to 360.

Order held out of equilibrium

Ilya Prigogine and Isabelle Stengers, *Order Out of Chaos*. Bantam, 1984. Prigogine received the Nobel Prize in Chemistry in 1977 for work on dissipative structures.

Order sustained by throughput

Erwin Schrödinger, *What Is Life?* Cambridge University Press, 1944.

Impedance matching

The maximum power transfer theorem, standard in electrical engineering, commonly attributed to Moritz von Jacobi, 1840.

Prediction from gaps

Dmitri Mendeleev, periodic table of 1869 and the predictions of 1871. Gallium identified by Lecoq de Boisbaudran, 1875; germanium by Winkler, 1886.

Affordance

James J. Gibson, *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.

Relational properties

Carlo Rovelli, *Relational Quantum Mechanics*. International Journal of Theoretical Physics 35 (8), 1996, 1637 to 1678.

Reality as relation

Carlo Rovelli, *Helgoland: Making Sense of the Quantum Revolution*. Riverhead, 2021.

Agreement across observers

Emily Adlam and Carlo Rovelli, *Information Is Physical: Cross-Perspective Links in Relational Quantum Mechanics*. Philosophy of Physics 1 (1), 2023, article 4.

Emergent strategy

adrienne maree brown, *Emergent Strategy: Shaping Change, Changing Worlds*. AK Press, 2017. Developed from the fiction of Octavia Butler and from practice in Black liberation organising.

Sankofa

Akan concept of retrieving from the past what remains valuable in order to move forward.

Falsification

Karl Popper, *The Logic of Scientific Discovery*. Hutchinson, 1959. First published as *Logik der Forschung*, 1934.

Alchemy into chemistry

Lawrence M. Principe, *The Secrets of Alchemy*. University of Chicago Press, 2013.

Nothing is created from nothing. Everything worth making is a recombination of components that already exist, and the quality of the recombination depends entirely on how deeply those components are understood. Order is never free: every made thing is held away from equilibrium by a flow that something or someone continuously pays for. And the object of making is not to disappear, nor to dominate, but to match: to find the impedance at which the made thing and the flows it sits inside transfer energy to each other instead of fighting.

A tree does this. Most buildings do not.

That gap is the work.

Please send your results, disagreements, criticisms, failures and observations.

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